MILITARY SPECIFICATION SHEET

ELECTRON TUBE, RECEIVING

TYPES 3Q4 AND 3V4

The complete requirements for procuring the electron tubes described herein shall consist of this document and the latest issue of Specification MIL-E-1.

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

DESCRIPTION: Power amplifier pentode

Outline	---	5-2 (EIA)
Base	---	E7-1
Envelope	---	T5-1/2
Cathode	---	Coated filament

Base connections:

Pin No.	---	1	2	3	4	5	6	7
Element								
3Q4	---	-f (+prl)	a	g1	g2	fct g3 (-prl)	a	+f
3V4	---	-f (+prl)	a	g2	nc	fct g3 (-prl)	g1	+f

ABSOLUTE-MAXIMUM RATINGS:

Parameter:	Ef	Eb	Ec1	Ec2	Ik	Alt
Unit:	Vdc	Vdc	Vdc	Vdc	mAdc	ft
Maximum:	1.6 or 3.2 (prl) (ser)	100	---	100	13	Note 1
Minimum:	1.2 or 2.4 (prl) (ser)	---	---	---	---	---
TEST CONDITIONS:	1.4 (prl)	90	-4.5	90	---	---

GENERAL:

Qualification - Required

(B) denotes changes

3Q4, 3V4
FSC 5960

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 1								
1268	Total grid current	2		1.0	II	Ic	0	-1.0	µAdc
1341	Power output (2)	2	Esig = 3.2 Vac; Ef = 1.1 Vdc (prl); Rp = 0.01 Meg	1.0	II	Po	140	---	mW
1256	Electrode current (anode)	-		1.0	II	Ib	6.5	12.5	mAdc
1251	Audio frequency noise and microphonics	3		1.0	II	ER	---	18	mV
(R) 1201	Short and discontinuity detection	-		0.4	II	---	---	---	---
	Quality conformance inspection, part 2								
1301	Filament current	-		---	---	If	88	112	mAdc
1256	Electrode current (screen grid)	-		---	---	Ic2	1.3	3.1	mAdc
1306	Transconductance	-		---	---	Sm	1,800	2,500	µmhos
1341	Power output (1)	-	Esig = 3.2 Vac; Rp = 0.01 Meg	---	---	Po	210	---	mW
1331	Direct-interelectrode capacitance	-							
	3Q4			---	---	Cg1-p	---	0.41	pF
	3V4					Cg1-p	---	.40	pF
	3Q4		Without shield			Cin	3.8	7.2	pF
	3V4					Cin	3.8	7.3	pF
	3Q4					Cout	2.4	5.6	pF
	3V4					Cout	2.2	5.4	pF
1031	Low-frequency vibration	-	Rp = 2,000 ohms	---	---	Ep	---	400	mVac
(B) 1211	Insulation of electrodes	-		---	---	---	---	---	---
(B) 1121	Base strain	-		---	---	---	---	---	---
(B) 2126	Glass strain	-		---	---	---	---	---	---
1105	Permanence of marking	-		---	---	---	---	---	---
	Quality conformance inspection, part 3								
1501	Intermittent life	-	Group A	---	---	---	---	---	---
---	Intermittent life-test end point (500 hours):								
1341	Power output (1)	-		---	---	Po	135	---	mW

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NOTES:

1. See "Reduced pressure (altitude) rating", and altitude, maximum peak voltage in the basic document.
2. This test to be performed at the conclusion of the holding period.
3. Ebb = Ecc2 = 135 Vdc; Ec1 = 0; Rg1 = 3.3 Meg; Rp = 0.27 Meg; Rg2 = 2 Meg (by-passed with a 0.5 μF capacitor to -f). Remove tube and adjust gain of amplifier for 50 mW output with Esig = 3.0 Vac.

Custodians:
Army - EL
Navy - EC
Air Force - 80

Review activities:
Air Force - 11, 70
DSA - ES

User activities:
Army - WC, AV
Navy - AS, OS, MC, CG, SH
Air Force - 19

Preparing activity - Navy - EC

Agent: DSA - ES

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